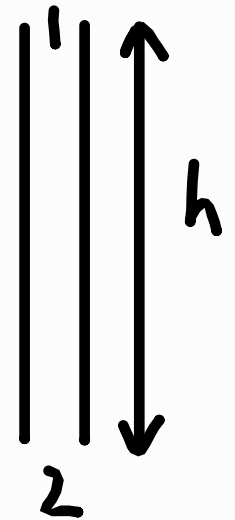


1a) viscosity of molten steel $= 2.6 \times 10^{-3} \text{ Pa s}$

The viscosity of molten steel is comparable to water, which has a viscosity of $1 \times 10^{-3} \text{ Pa s}$.

b) $Re = \frac{\rho V D}{\mu}$

Assuming no losses, energy equation from 1 to 2:



$$\cancel{\frac{P_1}{\rho g}} + \cancel{\frac{V_1^2}{2g}} + z_1 = \cancel{\frac{P_2}{\rho g}} + \frac{V_2^2}{2g} + \cancel{z_2}$$

$$h = \frac{V_2^2}{2g}$$

$$V_2 = \sqrt{2gh}$$

$$Re = \frac{\rho V D}{\mu} = \frac{\rho \sqrt{2gh} D}{\mu}$$

$$20000 = \frac{7850 \sqrt{2gh} (20 \times 10^{-3})}{2.6 \times 10^{-3}}$$

$$\sqrt{2gh} = 0.3312101911$$

$$h = 5.591243154 \times 10^{-3} \text{ m}$$

$$\approx 5.59 \text{ mm}$$

$$1c) Re = \frac{\rho \sqrt{2gh} D}{\mu}$$

$$2000 = \frac{7850 \sqrt{2gh} (20 \times 10^{-3})}{2.6 \times 10^{-3}}$$

$$\sqrt{2gh} = 0.03312101911$$

$$h = 5.591243134 \times 10^{-5} \text{ m}$$

$$\approx 55.91 \mu\text{m}$$

d) If there is friction, v will decrease, hence the minimum height will increase.

e) In practice, the pouring cup will be at least a few centimetres above the metal level in the mould. Thus, the flow is likely to be turbulent.

2a) During metal casting, hot liquid metal is poured into the cavity. During solidification, most metals will cool down and shrink. The shrinkage in the cavity will lead to the formation of a defect if nothing is done.

Thanks to the riser, we have a reservoir of liquid to fill up the gaps that might form due to shrinkage. The metal inside the riser should remain liquid longer than the cast part.

Therefore, the disadvantage of an open riser is that it might cool down faster due to contact with air. However, an advantage is that an open riser is easier to make than a blind riser.

b) Chvorinov's rule:

$$T_{Ts, cast} = C_m \left(\frac{V_{cast}}{A_{cast}} \right)^2$$

$$T_{Ts, riser} = C_m \left(\frac{V_{riser}}{A_{riser}} \right)^2$$

$$\begin{aligned} V_{cast} &= 12.5 \times 2.0 \times 7.5 \\ &= 187.5 \text{ cm}^3 \end{aligned}$$

$$\begin{aligned} A_{cast} &= 2 \times 12.5 \times 2.0 + 2 \times 12.5 \times 7.5 + 2 \times 7.5 \times 2.0 \\ &= 267.5 \text{ cm}^2 \end{aligned}$$

$$2b) T_{Ts, cast} = C_m \left(\frac{V_{cast}}{A_{cast}} \right)^2$$

$$1.6 = C_m \left(\frac{187.5}{267.5} \right)^2$$

$$C_m = 3.256604444$$

$$\approx 3.26 \text{ cm min}^{-1}$$

$$V_{riser} = \frac{\pi D^2}{4} D = \frac{\pi D^3}{4}$$

$$A_{riser} = \frac{\pi D^2}{4} \times 2 + \frac{2\pi D}{2} \times D$$

$$= \frac{\pi D^2}{2} + \pi D^2$$

$$= \frac{3}{2} \pi D^2$$

$$T_{Ts, riser} = C_m \left(\frac{\frac{\pi D^3}{4}}{\frac{3}{2} \pi D^2} \right)^2$$

$$2.0 = 3.26 \left(\frac{1}{36} D^2 \right)$$

$$D = 4.702012102$$

$$\approx 4.7 \text{ cm}$$

$$3) Q = \frac{V}{T} = A_v$$

$$V = A_v T$$

$$= AT\sqrt{2gh}$$

$$= 400 \times 10^{-6} \times 1.35 \times \sqrt{2 \times 9.81 \times 175 \times 10^{-3}}$$

$$= 1.00066418 \times 10^{-3} \text{ m}^3$$

$$\approx 1 \text{ dm}^3$$

4) No. Grey cast iron is an iron with high C content. When the metal cools graphitisation occurs, which means graphite flakes form, causing the metal to expand.

5) Dental crowns are too small to be produced using machining or forging. The surface finish also needs to be very good, so sandcasting is not ideal. Hence, dental crowns are made using investment casting.

An advantage of using investment casting to produce dental crowns is that they can be made exactly for the patient by making a copy of the patient's teeth initially.